

Action Methods

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Details with Examples

1. Default Action Methods

- General methods used to handle basic requests.

Example:

```
public IActionResult Index()
{
    return View();
}
```

2. HTTP Verb-Specific Action Methods

- Mapped to specific HTTP methods using attributes.

a. [HttpGet]

- Handles GET requests for retrieving data.

Example:

```
[HttpGet]
public IActionResult Details(int id)
{
    var model = _service.GetDetails(id);
    return View(model);
}
```

b. [HttpPost]

- Handles POST requests to submit data.

Example:

```
[HttpPost]
public IActionResult Create([FromBody] MyModel model)
{
    _service.Add(model);
    return Ok();
}
```

c. [HttpPut]

- Handles PUT requests for updating data.

Example:

```
[HttpPut("{id}")]
public IActionResult Update(int id, [FromBody] MyModel model)
{
    _service.Update(id, model);
    return NoContent();
}
```

d. [HttpDelete]

- Handles DELETE requests for deleting data.

Example:

```
[HttpDelete("{id}")]
public IActionResult Delete(int id)
{
    _service.Delete(id);
    return Ok();
}
```

-

e. [HttpPatch]

- Handles PATCH requests for partial updates.

Example:

```
[HttpPatch("{id}")]
public IActionResult UpdatePartial(int id, [FromBody]
JsonPatchDocument<MyModel> patchDoc)
{
    var model = _service.GetById(id);
    patchDoc.ApplyTo(model);
    return Ok(model);
}
```

3. Async Action Methods

- Improves scalability using asynchronous processing.

Example:

```
[HttpGet]
public async Task<IActionResult> GetItems()
{
    var items = await _service.GetAllAsync();
    return Ok(items);
}
```

4. API-Specific Action Methods

- Found in API controllers (`[ApiController]`).
- Return JSON or XML data.

Example:

```
[ApiController]
[Route("api/[controller]")]
public class MyApiController : ControllerBase
{
    [HttpGet]
    public IActionResult GetItems()
    {
        return Ok(new { id = 1, name = "John" });
    }
}
```

5. Custom Action Methods

- Use custom attributes or logic for special purposes.

Example:

```
[Route("custom")]
public IActionResult CustomLogic()
{
    return Content("Custom Logic Executed");
}
```

6. File-Related Action Methods

- Used to upload/download files.

File Upload:

Example:

```
[HttpPost]
public IActionResult UploadFile(IFormFile file)
{
    var path = Path.Combine(Directory.GetCurrentDirectory(),
"uploads", file.FileName);
    using var stream = new FileStream(path, FileMode.Create);
    file.CopyTo(stream);
    return Ok("File uploaded");
}
```

File Download:

Example:

```
[HttpGet]
public IActionResult DownloadFile()
{
    var fileBytes = System.IO.File.ReadAllBytes("file.txt");
    return File(fileBytes, "text/plain", "file.txt");
}
```

7. Redirect and Status-Result Methods

- Redirects and returns HTTP status codes.

Redirect Example:

```
public IActionResult RedirectToGoogle()
{
    return Redirect("https://www.google.com");
}
```

Status Result Example:

```
public IActionResult NotFoundExample()
{
    return NotFound("Resource not found");
}
```